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Emilio Ferrer, Ph.D.
Editor-in-Chief, *Multivariate Behavioral Research*

Dear Dr. Ferrer:

We are pleased to resubmit our revised manuscript, "From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models" (MBR-26-107), for consideration in Multivariate Behavioral Research.

Thank you and the Associate Editor and reviewers for the thoughtful assessment of our work. The comments helped us clarify the connection between idiographic modeling and meta-analytic SEM and strengthen the evaluation of MetaVAR. The revision examines zero and larger transition heterogeneity, compares two DSEM prior specifications, distinguishes sampling uncertainty from between-person variation, and adds an empirical comparison with the uncertainty-uncorrected approach.

We have also clarified the limits of the evidence. The paper discusses higher-dimensional estimation as a limitation, reports variance-boundary difficulties, and explains why the empirical DSEM fits do not support a reliable numerical comparison. The revised interpretation emphasizes MetaVAR's contribution to propagating uncertainty in person-specific estimates while acknowledging conditions in which DSEM performs better or simpler summaries give similar results.

The point-by-point responses below reproduce the reviewers' comments and identify the changes and their locations. Page references refer to the supplied manuscript.pdf and supplement.pdf, each numbered separately; they do not refer to the tracked manuscript. Reviewer 2's unnumbered comments are grouped into 16 items for ease of reference.

Thank you for considering the revised manuscript.

Sincerely,

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Response to the Associate Editor and Reviewers

Associate Editor

Comment AE-1

> 1) Both reviewers mentioned the magnitude of heterogeneity in parameters, but they had opposite recommendations (for different reasons). Reviewer 1 suggested using larger variances (to offset the impact of priors) and Reviewer 2 suggested smaller variances (to assess how noisy estimate would be; although reviewer two also mentioned that low but non-zero population values did make some simulation results less clear, similar to Reviewer 1). I think both points are compelling; my takeaway was that some additional justification for the population values might be helpful (possibly including additional text in the discussion to address how results may potentially change for conditions that were not studied).

Response. Thank you for bringing these complementary concerns together. The revised simulation now includes zero, reference, and doubled transition heterogeneity, with both default and alternative DSEM priors. We explain why the reference values were chosen and distinguish small nonzero variation from exact homogeneity. The expanded Results show how the methods respond to these conditions: MetaVAR's transition-variance coverage advantage is clearest at reference heterogeneity, while DSEM becomes more competitive as heterogeneity increases. We also acknowledge that varying heterogeneity around one mean dynamic system does not establish performance across all generating processes.

Location in the revision. Manuscript, pp. 14, 21, 32. Heterogeneity design, simulation findings, and scope of generalization.

Comment AE-2

> 2) Both reviewers also mentioned that performance with a larger number of variables may be of interest given the potential complications that may arise under these conditions. If this is outside of the paper's scope, the discussion should incorporate some text to outline how performance may possibly be affected by a larger number of variables.

Response. We have taken the discussion-based approach you suggested. The General Discussion now explains how the number of features and between-person covariance components grows with the number of variables, why both individual series length and participant sample size matter, and how covariance restrictions might help. We explicitly limit the performance conclusions to the bivariate evaluation. The computational advantage observed in the benchmark is not presented as evidence that an unrestricted higher-dimensional model will be estimable.

Location in the revision. Manuscript, pp. 32–33. Dimensionality and covariance restrictions in the General Discussion.

Comment AE-3

> 3) There are several comments suggesting potential clarifications. I think the idea to link these two frameworks is insightful, but a potential challenge of blending two different frameworks is that

> readers with expertise in MASEM may not necessarily be familiar with idiographic modeling and vice versa. As a global comment, during the revision, I would just suggest that the authors pay special attention to making sure that the paper is accessible to the broader audience that may be interested in this work, which may include readers who only have knowledge of one of methods being combined.

Response. Your accessibility concern helped us identify where the connection between the two frameworks needed to be clearer. The Stage-1 exposition now avoids or explicitly defines jargon that may have very different meanings in the MASEM and VAR frameworks (e.g., "heterogeneity parameters" as brought up by R2 as a point of confusion), and clearly distinguishes a person's estimated dynamic parameters, their sampling covariance, and the innovation covariance within that person's VAR. The Stage-2 exposition then explains howperson-specific estimates with uncertainty support inference about population means and between-person variation. We have also clarified the simulation targets before the Results and organized the findings around patterns, with numerical detail in the figures. These changes preserve the paper's existing progression.

Location in the revision. Manuscript, pp. 7–8, 18, 20. Stage 1 exposition, performance targets, and Results.

Reviewer 1

Comment R1-1

> 1. The simulation compares MetaVAR against a Bayesian multilevel VAR (BMLVAR) implemented in Mplus using default or "relatively uninformative" priors. The authors themselves note that "recovery of multilevel variance–covariance components can depend materially on prior specification, especially when heterogeneity parameters are small, weakly identified, or near the boundary" (p. 16). The DGP in this simulation has small heterogeneity variances for the transition parameters: the diagonal entries of Σ_A are 0.017, 0.008, 0.001, and 0.026 (Equation 11). These are exactly the conditions known to introduce substantial bias and undercoverage. This does not necessarily indicate MetaVAR is better than BMLVAR. The authors may need to consider other common priors in the BMLVAR to address this concern.

Response. We agree that the original comparison could not separate a method difference from sensitivity to priors near small variances. The revision therefore reports default and alternative DSEM priors throughout, together with doubled transition heterogeneity. The Methods give the alternative hyperparameters and explain their rationale; the supplement notes that the proper variance priors remain heavy-tailed and influential near zero.

The interpretation has changed accordingly. DSEM generally recovered transition means better, and its variance coverage became more competitive at larger heterogeneity. Alternative priors also improved covariance point recovery in many conditions. MetaVAR's contribution is framed around propagation of first-stage uncertainty and its demonstrated heterogeneity-recovery benefits, rather than superiority to Bayesian estimation in general. We also report posterior-quality counts so that normal termination is not mistaken for adequate mixing.

Don't understand what the red sentence means and its purpose. If you previously took "normal termination" as adequate mixing, that just means we failed at allowing for adequate burn-in. I don't understand what "posterior-quality counts" are, and how they help us get around the issue. We simply just needed to let the estimation run longer. Can you rephrase perhaps so it doesn't create a red flag? Better if you just say something like: Mixing quality and other convergence diagnostics were examined in greater depth to filter out cases showing dissatisfactory mixing and non-convergence.

We distinguish available DSEM summaries from satisfactory posterior sampling. Every required parameter had to satisfy the convergence, effective-sample-size, and Monte Carlo error criteria for a replication to pass. The primary results retain all usable runs, including those failing this screen. Consequently, the DSEM comparisons describe the evaluated priors and computational settings, rather than performance guaranteed to reflect fully converged posterior sampling.

Location in the revision. Manuscript, pp. 4, 17, 20–21, 30; Supplement, pp. 2, 6. DSEM priors and diagnostic criteria, usable-run reporting, and interpretation; supplement prior note and Table 1.

Comment R1-2

> 2. The entire simulation is conducted with k = 2 variables. For a bivariate VAR(1), the Stage 1 parameter vector θ_i has dimension p = 9 (two set points, four transition coefficients, three innovation parameters). The between-person covariance τ^2 is a 9 × 9 matrix, estimated from N ∈ {50, 100, 200}

persons. In real EMA applications, k $\geq$ 4 variables are common (e.g., distinct positive and negative affect facets, stress, arousal). It is unclear whether MetaVAR is practically estimable in higher-dimensional settings, and no guidance is provided. A related question is that only one set of population values (Eq. 11) was used, making it unclear how generalizable the findings are to other settings.

Response. We agree that the bivariate evaluation leaves an important question about practical estimability. We first clarified the dimensions actually fitted: Stage 1 estimated a nine person-specific parameters (features), but the simulation synthesizes six and uses separate set-point and transition blocks, with 13 heterogeneity components. The empirical model uses all nine features and 45 components.

The R wanted higher k, so your point about there already being 9 features (person-specific parameters, I think) doesn't help. You are better off restating the usefulness of the results despite a low k, and stating that this is a future direction we address in the Discussion section.

The General Discussion now addresses the demands of larger systems and the potential role of covariance restrictions, following the Associate Editor's suggestion to discuss this limitation. Zero and doubled heterogeneity broaden the simulation, but we explicitly acknowledge the remaining restrictions to one mean system, fixed innovation covariance, and correct specification. We do not claim demonstrated higher-dimensional performance.

Location in the revision. Manuscript, pp. 17, 32–33. Six-feature simulation specification and dimensionality limitations.

Comment R1-3

3. The between-person covariance τ^2 is a p $\times$ p matrix estimated from N persons. Near the boundary of the positive-semidefinite cone, standard ML confidence intervals lose their asymptotic validity. The paper does not report the rate of boundary solutions in the simulation, how near-zero diagonal elements are handled (are they fixed to zero? constrained?), or what the implications of near-boundary estimates are for reported coverage rates. Given that several diagonal elements of τ^2 under the DGP are small (e.g., the off-diagonal transition parameters have very small population variances), boundary estimation is plausible for smaller N and T cells.

Response. This was a consequential omission. The supplement now specifies the covariance parameterization, defines near-zero variances and near-singular matrices, and reports their frequencies by condition alongside usable replication counts. Near-boundary estimates are retained when otherwise usable; they are not silently discarded or fixed to zero.

The Results now make the inferential limitation explicit. At zero transition heterogeneity, normal-theory interval exclusion ranged from .004 to .398, and robust intervals did not resolve the problem. We distinguish near singularity from a near-zero diagonal and explain why Bayesian credible intervals do not perform satisfactorily as an inferential test for variance-related parameters that are close to 0. The Discussion also separates assuming a common effect from estimating heterogeneity at the boundary.

Location in the revision. Manuscript, pp. 21–23; Supplement, pp. 2, 6. Zero-heterogeneity findings and common-effect interpretation; supplement boundary parameterization and Table 1.

Comment R1-4

> 4. All nine I^2 values in the empirical illustration exceed 0.94, with several near 0.999. The authors present these as evidence that "the observed dispersion in Stage 1 parameter estimates was dominated by true between-person heterogeneity rather than sampling variability"(p. 25). However, when N = 217 persons each contribute relatively long series (up to 150 occasions after preprocessing), the typical Stage 1 sampling variance $\tilde{v}_p$ is small by construction, inflating I^2. This makes I^2 near unity an almost inevitable artifact of the sample characteristics rather than a substantive finding about affect dynamics. The paper does not discuss this problem; the I^2 metric, developed for traditional meta-analysis where the number of studies is small and sampling variances are large, has different interpretive properties when N is large and Stage 1 estimates are precise. Furthermore, the usefulness of I^2 as a measure of heterogeneity has been questioned in meta-analysis. Most meta-analysts recognize that I^2 is a relative measure of heterogeneity influenced by sample sizes.

Response. We agree that the earlier wording gave I^2 too much substantive weight. The revised text describes it as a measure relative to typical Stage 1 sampling variance, and explicitly notes that more precise individual estimates can produce a large I^2 without greater absolute heterogeneity. We now report the between-person standard deviations in addition to I^2 in the Results section and elaborate on the need for an improved I^2 index for intensive longitudinal models. The empirical discussion also distinguishes the information in observed reports from the length of the regularized time grid.

How does this point matter? Does considering one as opposed to the other gives a broader range of I^2. I think we just need to make sure to address this as a limitation and future direction in the Discussion section.

The revised illustration uses the final analytic sample of 212 participants.

Location in the revision. Manuscript, pp. 12, 26, 28, 42. Definition and interpretation of I-squared; empirical Table 1 (manuscript p. 42).

Comment R1-5

> 5. For the bivariate case, τ^2 is 9×9 with 45 free parameters estimated from N persons. As the number of psychological variables k increases, the parameter vector θ_i grows, and the number of free parameters in τ^2 grows hugely. For k = 5 variables (a modest EMA protocol), the Stage 1 parameter vector has dimension ˜55 and τ^2 contains 1,540 free parameters. This is clearly infeasible without strong structural constraints on τ^2. This issue is well-known in MASEM. For instance, the metaSEM package uses diagonal τ^2 as the default option to address this issue. However, the paper does not discuss how MetaVAR scales with k.

Response. We share this concern and have added an explicit account of covariance growth. For the feature vector used here, including the unique innovation-covariance elements gives 45 features at k = 5 and therefore 1,035 distinct heterogeneity components. Although these counts differ from the example in the comment, they support the same conclusion: unrestricted covariance estimation quickly becomes demanding. The Discussion now considers diagonal, block, and theory-selected restrictions, while identifying sparse and low-rank structures as future work. It also explains why parallelizing Stage 1 does not solve the information constraints at Stage 2.

Location in the revision. Manuscript, pp. 32–33. Parameter growth and covariance restrictions.

Comment R1-6

> 6. Only the MetaVAR was used in the empirical illustration. I suggest including the BMLVAR and uncertainty-uncorrected approaches in this illustration. The authors should discuss whether the findings from the simulation studies are also evident in the empirical illustration.

Response. The empirical comparison is much more informative with the uncorrected analysis included. We now present MetaVAR and Uncorr using the same nine-feature Stage 1 estimates. Set-point means are nearly identical, whereas inertia means and several dispersion estimates differ. This is consistent with uncertainty propagation affecting some features more than others, although the empirical data cannot establish which estimator is more accurate.

Following your suggestion, we attempted to add DSEM to the empirical comparison using both default and alternative priors. Despite approximately 36 hours of computation per fit—about three days of computation across the two fits—neither achieved adequate posterior convergence. We therefore cannot report a reliable numerical comparison with DSEM: the PA-inertia variance has rank-normalized R-hat values of 1.442 and 1.462 and bulk effective sample sizes below eight, and inertia means also show inadequate mixing. The manuscript reports this limitation and the approximately 36-hour runtimes; the supplement gives a brief diagnostic explanation, with full output and traces in the compendium. Thus, we have implemented the MetaVAR–Uncorr comparison and investigated DSEM, but cannot fully answer the requested DSEM comparison without better-converged fits. We have made that limitation explicit rather than interpreting unstable estimates.

Location in the revision. Manuscript, pp. 26, 28, 42; Supplement, pp. 2–3. Empirical comparison and diagnostic limitations; empirical Table 1 (manuscript p. 42).

I am concerned that you didn't use alternative priors that make things easier for DSEM. So this part sounds weak to me. To me you need to make the IW prior much more restricted (stronger faith, higher df) around the true values, with a base cov matrix that is diagonal. Did you already do that? Can we talk?

Reviewer 2

Comment R2-1

> Simulation study and heterogeneity parameters: As someone familiar with meta analysis in this framework, it is unclear how heterogeneity was quantified here in the data generating mode and evaluation measures. In general the results on heterogeneity parameters in the Monte Carlo study is very hard to read.

Response. The original terminology did not make the target clear enough. We now define heterogeneity in the data-generating process as the between-person covariance of the underlying set points and transition coefficients. The performance section identifies the six population means and 13 heterogeneity components being evaluated, and the Results distinguish mean recovery from variance and covariance recovery. This makes clear that the heterogeneity results concern population variance–covariance parameters, not the estimation of each person's realized random deviation.

Location in the revision. Manuscript, pp. 14, 18, 21. Definition of heterogeneity, recovery targets, and heterogeneity Results.

Comment R2-2

> The paper seems very interested in being able to detect heterogeneity in parameters. While some variances of parameters across individuals are close to zero (i.e., low heterogeneity in the data generating values), it was curious to see that no parameters were explicitly set to have no heterogeneity across people. This would help to assuage concern that homogenous parameters might look heterogeneous due to noise, large sample size, etc. in some modeling frameworks (Hoekstra, Epskamp, and Borsboom, 2023, is a nice motivator of this). While it's great to take inspiration from an empirical example, when the parameters are nearly zero (as in the variance across people) then perhaps it is safe to assume the population-generating parameters is zero, and generate you data accordingly.

Response. The revised simulation now includes an exact-zero transition-heterogeneity condition while retaining heterogeneous set points. This directly examines whether uncertainty in idiographic estimates can appear as between-person variation. ~~Uncorr continued to estimate positive transition variation; MetaVAR reduced that distortion, but its Wald intervals also showed important boundary limitations. In addition, we now report interval exclusion of zero separately from detection of nonzero effects.~~

I don't see why these sentences help address the R's main concern.

The supplement includes a separate zero-target figure, and the Methods explain why relative bias is undefined at zero.

How do you handle the 0 inclusion for the DSEM? Bayesian estimation cannot easily be set to have 0 variance even if the true variance value is indeed 0.

Location in the revision. Manuscript, pp. 14, 19, 21; Supplement, p. 12. Zero-heterogeneity design and inference; supplement Figure 6 (p. 12).

Comment R2-3

> Heterogeneity Parameters needs to be defined. It enters into the manuscript in the results section and isn't described in the description of measures. The Performance Measures is a bit vague. It would help to be clear exactly which parameters will be tested for coverage, power, etc. and the inferences made from that. It wasn't clear at first when reading the results why bias in random effects was in the "heterogeneous parameters" section.

Response. We have moved the definition into the Methods, before the results are introduced. "Heterogeneity parameters" now refers explicitly to the 13 freely estimated between-person variances and covariances of the six synthesized features. The performance section states which targets enter each metric: bias and RMSE assess point recovery, coverage assesses interval calibration, and detection assesses sensitivity to nonzero targets. It also distinguishes these population parameters from individual deviations.

Location in the revision. Manuscript, pp. 18–19. Performance Metrics and point-estimate measures.

Comment R2-4

> I took the heterogeneity parameters to be the I^ 2 index (as described on page 11). Yet when we get to the results, it seems that random effects covariances and variances are what is used directly (second sentence of this subsection) with bias and coverage for these estimates reported. It also seems like bias of the random effects was considered under "heterogeneity parameters". This is all very confusing. I saw no reference to I^2 anywhere in the paper after being introduced despite it initially being offered as an assessment of heterogeneity. I had expected bias in I^2 to be explored.

Response. Thank you for identifying this ambiguity. We now distinguish the heterogeneity covariance, which is a simulation recovery target, from I^2, which is used descriptively in the empirical illustration. The framework explicitly states that the simulation does not evaluate recovery of I^2. The empirical Results and Discussion explain its dependence on Stage 1 precision and accompany it with standard deviations in the original parameter units.

I think the key point you should say is acknowledging the inadequacy of the I^2 as an "effect size" quantifier in intensive longitudinal models/data, and suggesting some possible directions for future extensions in the Discussion section.

Location in the revision. Manuscript, pp. 12, 18, 26, 42. I-squared versus covariance recovery; empirical Table 1 (manuscript p. 42).

Comment R2-5

> Relatedly, some parameter values in the A matrix have very low variance across people. Some discussion of the interpretability of this is needed. It was hard to ascertain exactly what was done, but it seems like power to detect effects was taken variance estimates (in addition to random effects). Some variances across people in the A matrix are very small (.001) - if this is not detected as significant, doesn't that make sense?

Response. Yes—low detection of a very small nonzero variance need not be surprising or substantively troubling. We now explain that detection must be read alongside the target's magnitude, bias, RMSE, and coverage. The revised Results describe how detection changes with series length and heterogeneity and explain why higher uncorrected detection can be misleading when coverage is poor. Exact-zero targets are handled separately because inference at a variance boundary poses a different problem.

Location in the revision. Manuscript, pp. 20–22. Interpretation of detection for small nonzero and exact-zero targets.

Comment R2-6

> One benefit of two-stage random effects for VAR models is that you can include all covariates in the VAR model. Anecdotally (and in all tutorials on the method), DSEM seems to have a hard time estimating models with a larger number of variables than 2. It would strengthen the paper to show that MetaVAR can perform well with a larger set of variables.

Response. A larger-variable study would be valuable, but it is beyond the evidence established here. Following the Associate Editor's suggested alternative, we have added a substantive discussion of dimensionality: growth in the parameter vector and covariance matrix, the demands on both stages, and the assumptions introduced by covariance restrictions. The paper now states plainly that the bivariate evaluation does not demonstrate higher-dimensional performance.

Location in the revision. Manuscript, pp. 32–33. Higher-dimensional limitations and future work.

Comment R2-7

> A figure of power results would help. As a general rule if you have more that 10 values in a paragraph you should probably put it in a table or a figure. To compare "power; results, for instance, i had to highlight all uses of that word in the results section and then track the results. This is onerous for the reader, a figure would be much easier (or a table if you want to put all results together).

Response. We have made the detection results visible in the figures and reduced the need to track numbers through the prose. Main Figure 1 summarizes detection alongside coverage and zero-target exclusion; the supplement's detection figure shows the individual parameters and conditions, with zero-target exclusion in a separate figure. The Results now explain the effects of sample size, series length, and heterogeneity across the design, including where MetaVAR helps, where the methods are similar, and where DSEM performs better.

The R explicitly asked for power results. You said a lot of things here, but you didn't say the most critical point – that power results are now elaborated explicitly XXXX.

Location in the revision. Manuscript, pp. 20, 22, 43; Supplement, pp. 11–12. Detection Results; manuscript Figure 1 (p. 43) and supplement Figures 5–6 (pp. 11–12).

Comment R2-8

> Simulation studies 10-12 had heterogeneous T_i (i.e., some individuals had 50, some had 100, some had 200). I could not find discussion of the impact of this design on outcomes.

Response. We now describe both the allocation and what the unequal-length conditions show. Lengths cycle through 50, 100, and 200, with mean lengths of 115–116.25 depending on participant sample size. At matched N and heterogeneity, MetaVAR and DSEM generally maintained coverage close to their balanced-length counterparts, whereas Uncorr generally deteriorated. MetaVAR's heterogeneity RMSE improved in 80 of 117 comparisons with balanced T = 100, versus 10 for Uncorr. We also state the important qualification: the unequal-length conditions contain more observations on average, so this comparison does not isolate unequal precision at a fixed information budget.

Location in the revision. Manuscript, pp. 16, 22. Unequal-length allocation and condition effects.

Comment R2-9

> Page 7: when saying "In Stage 1, each individual .... Is fit with an idiographic dynamic model" perhaps specify that this is done outside of the random effects model in SEM. As in, it can be estimated with any VAR package of their choosing.

Response. We have clarified that the idiographic models are fitted separately, outside the Stage 2 random-effects SEM. Stage 2 can accept estimates from another suitable estimator, provided it supplies compatible parameter definitions and their aligned joint sampling covariance. This requirement is important: a list of coefficients and standard errors alone does not preserve the cross-parameter uncertainty that MetaVAR is designed to carry forward.

Location in the revision. Manuscript, pp. 7–8. Separate Stage 1 estimation and required outputs.

Comment R2-10

> When describing the empirical analysis you describe that you use the A matrix and covariance matrix of residuals as inputs to the meta-analytic framework. What about the sampling covariance matrix that quantifies estimation uncertainty?

Response. You are right: the earlier description omitted a central input. The empirical section now states that each participant contributes both the nine-feature estimate vector and its full 9-by-9 sampling covariance matrix. This is distinct from the innovation covariance within the VAR. We checked the transfer against the saved Stage 1 fits, and the supplement briefly records the numerical conditioning applied to the sampling covariances. The revised account makes explicit where first-stage uncertainty enters the synthesis.

Location in the revision. Manuscript, pp. 25–26; Supplement, p. 2. Empirical Stage 2 inputs and sampling-covariance conditioning.

Comment R2-11

> Sometimes BMLVAR is used, sometimes DSEM, for what seems to be the same one. DSEM is the more popular term, so it makes sense to use that throughout.

Response. We have standardized the manuscript and figure terminology to DSEM and defined it when introduced. This avoids suggesting that BMLVAR and DSEM denote two different comparison methods.

Location in the revision. Manuscript, pp. 2, 17. DSEM terminology introduced in the Abstract and specified in Methods (manuscript p. 17).

Comment R2-12

> Some rationale for using fitVARMxID is needed. Why use this when other approaches (such as MIIV-2SLS) have been shown to provide maximum-likelihood equivalent estimates in the SEM framework? Why not use Kalman Filter estimation (i.e., the vars package)? Is MetaVAR flexible and the user can choose? (even though simulation results are grounded to this package)?

Response. We have added the rationale for using the fitVARMxID R package. Its centered, likelihood-based VAR workflow provides the parameter vector and full joint sampling covariance needed for Stage 2, including missing-data handling. The same passage explains that MetaVAR is not conceptually tied to this package: another estimator can be used if it supplies compatible outputs. We limit the simulation conclusions to the implementation evaluated here and do not assume that alternative estimators or packages provide equivalent estimates or uncertainty information.

We should say why not MIIV-2SLS though. It doesn't give the parameter sampling covariance matrix at convergence, does it? Say something explicitly like that and address why not use MIIV-2SLS.

Location in the revision. Manuscript, p. 8. fitVARMxID R package rationale and flexibility of Stage 1.

Comment R2-13

> In the Performance Measures section it would be helpful to say how each one will be used to expectations that are clearly laid out in the preceding subsection.

Response. The performance section now connects each measure to the preceding expectations: point recovery is assessed with bias and RMSE, uncertainty calibration with coverage, and sensitivity to nonzero effects with detection. The Results follow these distinctions. They show why propagating uncertainty helps recover transition heterogeneity relative to Uncorr, while also recognizing DSEM's stronger recovery of transition means and the limits of Wald inference at zero.

Location in the revision. Manuscript, pp. 18, 20. Comparative expectations, performance measures, and Results.

Comment R2-14

> Tau-squared is reported for the empirical study - perhaps add to the simulation study. What are the true values in the data generating parameters? The obtained values?

Response. We now identify the simulation heterogeneity targets explicitly and distinguish the displayed generating covariance from the calibrated recovery targets after stability rejection. The Results explain recovery of the transition variances and covariances, and the supplement presents the full condition-level bias, RMSE, and coverage figures. For direct inspection of true and obtained values, the compendium file validated-performance.csv contains the target values, mean estimates, and Monte Carlo errors for every

parameter, condition, and method. Its location is .setup/notes/manuscript/MBR-26-107-R1/revision-execution/evidence/validated-performance.csv. We provide this complete numerical record electronically rather than reproduce hundreds of rows in the supplement, so readers can inspect individual targets without obscuring the main patterns.

Location in the revision. Manuscript, pp. 14, 21; Supplement, pp. 4, 8–10. Generating covariance and recovery Results; supplement figure guide and Figures 2–4 (pp. 8–10).

Comment R2-15

> I may have missed it, but was there a discussion of how the random effect estimates may differ from the individual-level estimates? That is, are those thought to be improved upon over solely individual-level analysis

Response. We have added this distinction to the empirical Discussion. MetaVAR's population means and heterogeneity covariances summarize the distribution of person-specific parameters; they do not replace the unpooled Stage 1 estimates. We therefore do not claim improved person-level estimation through shrinkage. Establishing such an improvement would require a separate evaluation of individual predictions.

Location in the revision. Manuscript, p. 29. Population synthesis versus person-level estimation.

Comment R2-16

> Is Stage 1 in the Uncertainty-Uncorrected Two-Stage Benchmark identical to Stage 1 in the MetaVAR, with the difference being that uncertainty isn't considered in UUTSB? Please clarify.

Response. Yes. The revised Methods now state that MetaVAR and Uncorr use the same saved Stage 1 fits, the same six point estimates, and the same block restrictions in the simulation. The difference is that MetaVAR carries their sampling covariance into Stage 2. The empirical comparison likewise uses identical nine-feature point estimates. This makes the comparison one of uncertainty propagation as a whole; it does not isolate the contribution of the off-diagonal sampling covariances.

Location in the revision. Manuscript, pp. 18, 26, 42; Supplement, p. 2. Common Stage 1 inputs; empirical Table 1 (manuscript p. 42).